



Childhood Pleuropulmonary Blastoma Treatment (PDQ®)–Health Professional Version

Types of Pleuropulmonary Blastoma

Pleuropulmonary blastoma is a rare and highly aggressive pulmonary malignancy that can present as a pulmonary or pleural mass. In most cases, pleuropulmonary blastoma is associated with germline pathogenic variants of the *DICER1* gene. The [International Pleuropulmonary Blastoma/DICER1 Registry](#) is a valuable resource for information about this rare malignancy.[1,2]

The following subtypes of pleuropulmonary blastoma have been identified:

Type I

Type I pleuropulmonary blastoma is a purely lung cystic neoplasm with subtle malignant changes that typically occurs in the first 2 years of life. Patients have a good prognosis. The median age at diagnosis for Type I tumors is 7 months,[3] and there is a slight male predominance. Transition from Type I to Type III occurs. However, a significant proportion of Type I lesions may not progress to Type II and Type III tumors.[2,4]

Histologically, these tumors appear as a multilocular cyst with variable numbers of primitive mesenchymal cells beneath a benign epithelial surface. Skeletal differentiation occurs in one-half of the cases.[4] This form of disease can be clinically and pathologically deceptive because of its resemblance to some developmental lung cysts, with over 10% discordance between local and central pathology review.[3]

Type Ir

Type Ir was originally recognized in older siblings of patients with pleuropulmonary blastoma but can also be seen in very young children. A lung cyst in an older individual with a *DICER1* variant or in a relative of a patient with pleuropulmonary blastoma is most likely to be Type Ir. [2] This is a purely cystic tumor that lacks a primitive cell component. The **r** designation signifies regression or nonprogression. In the National Cancer Institute Natural History of *DICER1* Syndrome study and the International Pleuropulmonary Blastoma/*DICER1* Registry, people with germline *DICER1* pathogenic or likely pathogenic variants who did not have computed tomography (CT) scans before age 12 years were screened with chest CT. Cystic

lesions were identified in 38% of individuals (42 of 110). Five cysts were resected, four of which were classified as Type Ir. The other cysts were not biopsied to confirm the Type Ir diagnosis.^[5]

In the International Pleuropulmonary Blastoma Registry, most Type I and Type Ir cysts are unilateral (74%), one-half are unifocal, and 55% are larger than 5 cm. Pneumothorax may be present at diagnosis in up to 30% of Type I and Type Ir pleuropulmonary blastoma cases.^[2]

Type II

In the International Pleuropulmonary Blastoma Registry, the median age at diagnosis is 35 months, and distant metastases are present at the time of diagnosis in 7% of patients.^[2]

Type II exhibits both cystic and solid components. The solid areas have mixed blastomatous and sarcomatous features. Most of the cases exhibit rhabdomyoblasts, and nodules with cartilaginous differentiation are common.^[6] Anaplasia is present in up to 60% of the cases.^[7]

Type III

In the International Pleuropulmonary Blastoma Registry, the median age at diagnosis is 41 months, and distant metastases are present at the time of diagnosis in 10% of patients.^[2]

Type III is a purely solid neoplasm, with the blastomatous and sarcomatous elements described for Type II. Anaplasia is present in 70% of patients.^[7-9]

In one report, 15 of 16 pleuropulmonary blastoma tumors were positive for IGF1R expression by immunohistochemistry.^[10] Genomic profiling showed amplification of the *IGF1R* gene in 4 of 16 pleuropulmonary blastoma tumors. All of these gene-amplified tumors were Type III.

The International Pleuropulmonary Blastoma Registry reported on 350 centrally reviewed and confirmed cases of pleuropulmonary blastoma over a 50-year period (see [Table 1](#)).^[2]

Table 1. Relative Proportions and Features of Pleuropulmonary Blastoma^a

	Type I	Type Ir	Type II	Type II/III or III
Relative proportion of pleuropulmonary blastoma cases	33%		35%	32%
Presence of germline <i>DICER1</i> pathogenic variant	75%	83%	63%	75%
Median age at diagnosis (months)	7	31	35	41
5-year overall survival rate	98%	100%	71%	53%
^a Adapted from Messinger et al.[2] and Nelson et al.[3]				

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Risk Factors and Surveillance

Germline *DICER1* Pathogenic Variants

Close to two-thirds of patients with pleuropulmonary blastoma have a germline *DICER1* pathogenic variant. Approximately one-third of families of children with pleuropulmonary blastoma manifest a number of dysplastic and/or neoplastic conditions comprising the *DICER1* syndrome.[\[1-3\]](#)

Germline *DICER1* pathogenic variants have been associated with the following:[\[1-5\]](#)

- Cystic nephroma and Wilms tumor. Up to 10% of patients with pleuropulmonary blastoma may develop cystic nephroma or Wilms tumor, which are the most relevant associated malignancies. These tumors are also more prevalent among family members.[\[6\]](#)
- Ovarian sex cord–stromal tumors (especially Sertoli-Leydig cell tumor).
- Multinodular goiter and thyroid carcinoma.[\[7\]](#)
- Uterine cervix embryonal rhabdomyosarcoma.
- Nasal chondromesenchymal hamartoma.
- Renal sarcoma.
- Pulmonary sequestration.
- Juvenile intestinal polyps.
- Ciliary body medulloepithelioma.

- Medulloblastoma.
- Pineoblastoma.
- Pituitary blastoma.
- Seminoma.

The penetrance of *DICER1* germline pathogenic variants associated with each pathological condition is not well understood, but lung cysts, pleuropulmonary blastoma, and thyroid nodules are the most commonly reported manifestations in individuals who have loss-of-function variants.[5] Most associated conditions occur in children younger than 10 years, although ovarian tumors and multinodular goiters are described in children and adults aged up to 30 years.[3,5] A study of 102 individuals with *DICER1* germline pathogenic variants revealed a neoplasm risk of 5% by the age of 10 years and 19% by the age of 50 years.[8] Surveillance and screening recommendations have been proposed.[5]

Surveillance

As with other cancer predisposition conditions, before individuals with *DICER1* pathogenic variants are screened, factors that must be considered include typical age of onset of each disease, potential benefits of early detection, and risks and availability of screening modalities. A consensus panel convened by the International Pleuropulmonary Blastoma Registry has proposed guidelines for surveillance. In addition to imaging-based surveillance, individuals and families can be counseled at each visit regarding potential signs and symptoms of *DICER1*-associated conditions and undergo appropriate age- and sex-specific preventive screening studies (see [Table 2](#)).[5]

Table 2. Potential Signs and Symptoms and Suggested Imaging Surveillance by System for Individuals With Germline *DICER1* Pathogenic Variants^a

System	Associated Condition	Signs/Symptoms to Consider	Screening: Clinical and Radiographic
Central nervous system and head and neck (excluding thyroid)	Macrocephaly	Pineoblastoma: Headache, emesis, diplopia, decreased ability for upward gaze, altered gait	Physical examination
	Pineoblastoma	Precocious puberty	Annual routine dilated ophthalmologic examination (generally unsedated) with visual acuity screening from age 3 years through at least age 10 years
	Pituitary blastoma	Pituitary blastoma: Cushing syndrome	Further testing if clinically indicated
	CBME	CBME: Decreased visual acuity and leukocoria	Recommend urgent MRI for any symptoms of intracranial pathology
	NCMH	NCMH: Nasal obstruction	

CBME = ciliary body medulloepithelioma; CT = computed tomography; CXR = chest x-ray; ERMS = embryonal rhabdomyosarcoma; MRI = magnetic resonance imaging; NCMH = nasal chondromesenchymal hamartoma; PPB = pleuropulmonary blastoma; SLCT = Sertoli-Leydig cell tumor; US = ultrasonography.

^aAdapted from Schultz et al.[5]

^bWhen CT is performed, techniques to minimize radiation exposure should be used. As novel MRI techniques are developed that will eventually allow detection of small cystic lesions, transition to nonradiation containing cross-sectional imaging should be considered.

System	Associated Condition	Signs/Symptoms to Consider	Screening: Clinical and Radiographic
Thyroid	Multinodular goiter	Visible or palpable thyroid nodule(s)	Baseline thyroid US by age 8 years, then every 3 years or with symptoms/findings on physical examination
		Persistent cervical lymphadenopathy	
	Differentiated thyroid cancer	Hoarseness	With anticipated chemotherapy or radiation therapy: Baseline US and then annually for 5 years, decreasing to every 2–3 years if no nodules are detected
		Dysphagia	
		Neck pain	
Lung	PPB	Tachypnea	CXR at birth and every 4–6 months until age 8 years, every 12 months at age 8–12 years; consider a chest CT ^b at age 3–6 months
	Lung cysts	Cough	Toddlers, if initial CT normal: Repeat between age 2.5 and 3 years

CBME = ciliary body medulloepithelioma; CT = computed tomography; CXR = chest x-ray; ERMS = embryonal rhabdomyosarcoma; MRI = magnetic resonance imaging; NCMH = nasal chondromesenchymal hamartoma; PPB = pleuropulmonary blastoma; SLCT = Sertoli-Leydig cell tumor; US = ultrasonography.

^aAdapted from Schultz et al.[5]

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System	Associated Condition	Signs/Symptoms to Consider	Screening: Clinical and Radiographic
	Pulmonary blastoma	Fever	If variant detected at age >12 years, consider baseline CXR or chest CT
		Pain	
		Pneumothorax	
Gastrointestinal	Small intestine polyps	Signs of intestinal obstruction	Education regarding symptoms recommended
Renal	Wilms tumor	Abdominal or flank mass and/or pain	Abdominal US every 6 months until age 8 years, then every 12 months until age 12 years
	Renal sarcoma		
	Cystic nephroma	Hematuria	If variant detected at age >12 years, consider baseline abdominal US
Female reproductive tract	SLCT	Hirsutism	For females beginning at age 8–10 years: Pelvic and abdominal US every 6–12 months at least until age 40 years
	Gynandroblastoma	Virilization	End of interval is undetermined, but

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^aAdapted from Schultz et al.[5]

^bWhen CT is performed, techniques to minimize radiation exposure should be used. As novel MRI techniques are developed that will eventually allow detection of small cystic lesions, transition to nonradiation containing cross-sectional imaging should be considered.

System	Associated Condition	Signs/Symptoms to Consider	Screening: Clinical and Radiographic
			current oldest patient with <i>DICER1</i> -associated SLCT was aged 61 years
	Cervical ERMS	Abdominal distension, pain, or mass	Education regarding symptoms strongly recommended

CBME = ciliary body medulloepithelioma; CT = computed tomography; CXR = chest x-ray; ERMS = embryonal rhabdomyosarcoma; MRI = magnetic resonance imaging; NCMH = nasal chondromesenchymal hamartoma; PPB = pleuropulmonary blastoma; SLCT = Sertoli-Leydig cell tumor; US = ultrasonography.

^aAdapted from Schultz et al.[5]

^bWhen CT is performed, techniques to minimize radiation exposure should be used. As novel MRI techniques are developed that will eventually allow detection of small cystic lesions, transition to nonradiation containing cross-sectional imaging should be considered.

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Clinical Presentation and Diagnostic Evaluation

Presenting symptoms for children with pleuropulmonary blastoma are not specific. They commonly include the following:

- Respiratory distress.
- Fever.
- Chest pain.

The tumor is usually located in the lung periphery, but it may be extrapulmonary with involvement of the heart/great vessels, mediastinum, diaphragm, and/or pleura.[1,2] Tumor embolism is a known risk, and radiographic evaluation of the central circulation is performed to identify potentially fatal embolic complications.[3]

Imaging evaluation may include chest radiography, computed tomography, magnetic resonance imaging, and echocardiography. Primary, recurrent, and/or extracranial metastatic pleuropulmonary blastoma presents with an fluorine F 18-fludeoxyglucose-avid lesion on positron emission tomography imaging.[4]

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Prognostic Factors

In a comprehensive analysis of 350 patients reported by the International Pleuropulmonary Blastoma Registry, only two prognostic factors were identified: the type of pleuropulmonary blastoma and the presence of metastatic disease at diagnosis.^[1] In three additional small cohort series, the ability to perform a complete surgical resection was also identified as a prognostic factor.^[2-4]

The presence of a germline *DICER1* pathogenic variant is not a prognostic factor.^[1]

A retrospective study analyzed TP53 expression by immunohistochemistry (IHC) in patients with pleuropulmonary blastoma.^[5] A total of 143 cases were included in the study, with the following distribution of pleuropulmonary blastoma types: Type I, 23%; Type Ir, 14%; Type II, 32%; and Type III, 31%. TP53 expression was determined by IHC and grouped as follows: 0%, 1% to 25%, 26% to 75%, and 76% to 100%. All Type I pleuropulmonary blastomas showed TP53 expressions of 0% to 25%, compared with Type III pleuropulmonary blastomas, which had higher TP53 expressions (>25%) ($P < .0001$). High TP53 expression (staining observed in >25% of the tumor cells) was significantly associated with age older than 1 year ($P = .0033$), neoadjuvant therapy ($P = .0009$), positive resection margin ($P = .0008$), and anaplasia ($P < .0001$). TP53 expression was significantly associated with recurrence-free survival ($P < .0001$) and overall survival ($P = .0350$). Higher TP53 expression was associated with a worse prognosis. Comparisons of concordance statistics showed no significant difference in prognostication when using morphological types compared with TP53 expression groups ($P = .647$).^[5]

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Treatment of Childhood Pleuropulmonary Blastoma

There are no standard treatment options for childhood pleuropulmonary blastoma. Current treatment regimens for these rare tumors have been informed by consensus opinion. The European Cooperative Study Group for Pediatric Rare Tumors within the PARTNER project (Paediatric Rare Tumours Network–European Registry) published comprehensive recommendations for the diagnosis and treatment of pleuropulmonary blastoma in children and adolescents.[\[1\]](#)

Treatment options for childhood pleuropulmonary blastoma include the following:

1. Surgery.
2. Adjuvant chemotherapy.

A complete surgical resection is required for cure.[\[2\]](#)

Data from the International Pleuropulmonary Blastoma Registry and the European Cooperative Study Group for Pediatric Rare Tumors suggest that adjuvant chemotherapy may reduce the risk of recurrence.[\[3\]](#); [\[4\]](#)[\[Level of evidence C1\]](#) Responses to chemotherapy have been reported with agents similar to those used for the treatment of rhabdomyosarcoma.[\[3-5\]](#)

Some general treatment considerations from the International Pleuropulmonary Blastoma Registry, according to subtype, are discussed below.[\[3,6\]](#)

Type I and Type Ir

Surgery is the treatment of choice for patients with Type I and Type Ir pleuropulmonary blastoma. In the International Pleuropulmonary Blastoma Registry series, the 5-year disease-free survival (DFS) and overall survival (OS) rates were 90% and 98%, respectively, for Type I, and 96% and 100%, respectively, for Type Ir. Approximately 10% of cases progressed to Type II or Type III after surgery. However, adjuvant chemotherapy has been used in almost 40% of patients with Type I disease, and it may be useful for patients at increased risk of recurrence or progression.[\[3,4,7\]](#)

The International Pleuropulmonary Blastoma/*DICER1* Registry reported that between 2006 and 2022, there were 205 children who had centrally reviewed Type I or Type Ir pleuropulmonary blastoma. Of these children, 39% with Type I and 5% with Type Ir received chemotherapy.[7]

- Patient outcomes were favorable, although 11 children (9 with Type I and 2 with Type Ir) experienced progression to Type II or III (n = 8) or regrowth of Type I at the surgical site (n = 3). None of these 11 children received chemotherapy before progression.
- The combination of age and cyst size was more useful than either factor alone in predicting whether a particular lesion was Type I or Type Ir.

Type II and Type III

For patients with Type II and Type III pleuropulmonary blastoma, a multimodal sarcoma treatment approach is recommended. This approach usually includes rhabdomyosarcoma regimens and either upfront or delayed surgery.[3,4,8] Anthracycline-containing regimens appear to be the most effective.[4]

In the Pleuropulmonary Blastoma Registry series, the 5-year DFS and OS rates were 59% and 71%, respectively, for Type II, and 37% and 53%, respectively, for Type III.[3] In this group of patients, approximately 50% of relapses occurred in the brain.[3]

The International Pleuropulmonary Blastoma/*DICER1* Registry reported the outcomes of children with Type II and Type III pleuropulmonary blastoma whose first treatment was ifosfamide, vincristine, dactinomycin, and doxorubicin (IVADo). From 1987 to 2021, 314 children with centrally confirmed Type II and Type III pleuropulmonary blastoma who received upfront chemotherapy were enrolled, 132 of whom (75 with Type II and 57 with Type III) received IVADo chemotherapy.[9]

- Adjusted analyses suggested improved OS for children treated with IVADo, compared with historical controls, with an estimated hazard ratio (HR) of 0.65 (95% confidence interval [CI], 0.39–1.08).
- Compared with localized disease, distant metastasis at diagnosis was associated with worse pleuropulmonary blastoma event-free survival and OS, with HRs of 4.23 (95% CI, 2.42–7.38) and 4.69 (95% CI, 2.50–8.80), respectively.

The role of radiation therapy is not well defined. While the use of radiation did not impact survival in the International Pleuropulmonary Blastoma Registry series, only 20% of patients with Type II and Type III received it.[3]

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Treatment of Progressive or Recurrent Pleuropulmonary Blastoma

A retrospective review included 35 children with Type II or Type III pleuropulmonary blastoma and progressive or recurrent disease who were registered in national and European databases and trials (2000–2018).^[1] Patients had a median age of 3.9 years (range, 0.5–17.8 years).

- The median time to progression was 0.58 years (range, 0.02–1.27 years) from diagnosis despite surgery, chemotherapy (n = 9), and radiation therapy (n = 1). All of these patients died.

- Patients were diagnosed with recurrent disease at a median age of 4.3 years (range, 1.7–15.1 years) and had a median delay to relapse of 1.03 years (range, 0.03–2.95 years).
- Recurrent disease occurred locally (n = 12), in combined sites (locally and metastatic) (n = 1), and in metastatic sites (n = 13), including the central nervous system (n = 11) and unspecified sites (n = 2).
- The 5-year event-free survival rate and overall survival (OS) rates for patients with recurrent disease were both 37% ($\pm$ 19%; 95% confidence interval).
- Local therapy (surgery and radiation therapy) had a favorable impact on OS (P = .03 and .02, respectively).

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Treatment Options Under Clinical Evaluation for Childhood Pleuropulmonary Blastoma

Information about National Cancer Institute (NCI)–supported clinical trials can be found on the [NCI website](#). For information about clinical trials sponsored by other organizations, see the [ClinicalTrials.gov website](#).

Special Considerations for the Treatment of Children With Cancer

Cancer in children and adolescents is rare, although the overall incidence has slowly increased since 1975.[1] Children and adolescents with cancer should be referred to medical centers that have a multidisciplinary team of cancer specialists with experience treating the cancers that occur during childhood and adolescence. This multidisciplinary team approach incorporates the skills of the following pediatric specialists and others to ensure that children receive treatment, supportive care, and rehabilitation to achieve optimal survival and quality of life:

- Primary care physicians.
- Pediatric surgeons.
- Pathologists.

- Pediatric radiation oncologists.
- Pediatric medical oncologists and hematologists.
- Ophthalmologists.
- Rehabilitation specialists.
- Pediatric oncology nurses.
- Social workers.
- Child-life professionals.
- Psychologists.
- Nutritionists.

For specific information about supportive care for children and adolescents with cancer, see the summaries on [Supportive and Palliative Care](#).

The American Academy of Pediatrics has outlined guidelines for pediatric cancer centers and their role in the treatment of children and adolescents with cancer.[2] At these centers, clinical trials are available for most types of cancer that occur in children and adolescents, and the opportunity to participate is offered to most patients and their families. Clinical trials for children and adolescents diagnosed with cancer are generally designed to compare potentially better therapy with current standard therapy. Other types of clinical trials test novel therapies when there is no standard therapy for a cancer diagnosis. Most of the progress in identifying curative therapies for childhood cancers has been achieved through clinical trials. Information about ongoing clinical trials is available from the [NCI website](#).

Dramatic improvements in survival have been achieved for children and adolescents with cancer. Between 1975 and 2020, childhood cancer mortality decreased by more than 50%.[3-5] Childhood and adolescent cancer survivors require close monitoring because side effects of cancer therapy may persist or develop months or years after treatment. For information about the incidence, type, and monitoring of late effects in childhood and adolescent cancer survivors, see [Late Effects of Treatment for Childhood Cancer](#).

Childhood cancer is a rare disease, with about 15,000 cases diagnosed annually in the United States in individuals younger than 20 years.[6] The U.S. [Rare Diseases Act of 2002](#) defines a rare disease as one that affects populations smaller than 200,000 people in the United States. Therefore, all pediatric cancers are considered rare.

The designation of a rare tumor is not uniform among pediatric and adult groups. In adults, rare cancers are defined as those with an annual incidence of fewer than six cases per 100,000 people. They account for up to 24% of all cancers diagnosed in the European Union and about

20% of all cancers diagnosed in the United States.[7,8] In children and adolescents, the designation of a rare tumor is not uniform among international groups, as follows:

- A consensus effort between the European Union Joint Action on Rare Cancers and the European Cooperative Study Group for Rare Pediatric Cancers estimated that 11% of all cancers in patients younger than 20 years could be categorized as very rare. This consensus group defined very rare cancers as those with annual incidences of fewer than two cases per 1 million people. However, three additional histologies (thyroid carcinoma, melanoma, and testicular cancer) with incidences of more than two cases per 1 million people were also included in the very rare group due to a lack of knowledge and expertise in the management of these tumors.[9]
- The Children's Oncology Group defines rare pediatric cancers as those listed in the International Classification of Childhood Cancer subgroup XI, which includes thyroid cancers, melanomas and nonmelanoma skin cancers, and multiple types of carcinomas (e.g., adrenocortical carcinomas, nasopharyngeal carcinomas, and most adult-type carcinomas such as breast cancers and colorectal cancers).[10] These diagnoses account for about 5% of the cancers diagnosed in children aged 0 to 14 years and about 27% of the cancers diagnosed in adolescents aged 15 to 19 years.[4]

Most cancers in subgroup XI are either melanomas or thyroid cancers, with other cancer types accounting for only 2% of the cancers diagnosed in children aged 0 to 14 years and 9.3% of the cancers diagnosed in adolescents aged 15 to 19 years.

These rare cancers are extremely challenging to study because of the relatively few patients with any individual diagnosis, the predominance of rare cancers in the adolescent population, and the small number of clinical trials for adolescents with rare cancers.

Information about these tumors may also be found in sources relevant to adults with cancer.

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Latest Updates to This Summary (04/03/2025)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Types of Pleuropulmonary Blastoma

Added [text](#) to state that in the National Cancer Institute Natural History of *DICER1* Syndrome study and the International Pleuropulmonary Blastoma/*DICER1* Registry, people with germline *DICER1* pathogenic or likely pathogenic variants who did not have computed tomography (CT) scans before age 12 years were screened with chest CT. Cystic lesions were identified in 38% of individuals. Five cysts were resected, four of which were classified as Type Ir. The other cysts were not biopsied to confirm the Type Ir diagnosis (cited Nelson et al. as reference 5).

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found on the [About This PDQ Summary](#) and [PDQ® Cancer Information for Health Professionals](#) pages.

About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the treatment of childhood pleuropulmonary blastoma. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

Reviewers and Updates

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Board members review recently published articles each month to determine whether an article should:

- be discussed at a meeting,
- be cited with text, or
- replace or update an existing article that is already cited.

Changes to the summaries are made through a consensus process in which Board members evaluate the strength of the evidence in the published articles and determine how the article should be included in the summary.

The lead reviewers for Childhood Pleuropulmonary Blastoma Treatment are:

- Denise Adams, MD (Children's Hospital Boston)
- Karen J. Marcus, MD, FACR (Dana-Farber of Boston Children's Cancer Center and Blood Disorders Harvard Medical School)
- William H. Meyer, MD
- Paul A. Meyers, MD (Memorial Sloan-Kettering Cancer Center)

- Thomas A. Olson, MD (Aflac Cancer and Blood Disorders Center of Children's Healthcare of Atlanta - Egleston Campus)
- Arthur Kim Ritchey, MD (Children's Hospital of Pittsburgh of UPMC)
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Levels of Evidence

Some of the reference citations in this summary are accompanied by a level-of-evidence designation. These designations are intended to help readers assess the strength of the evidence supporting the use of specific interventions or approaches. The PDQ Pediatric Treatment Editorial Board uses a [formal evidence ranking system](#) in developing its level-of-evidence designations.

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Updated: April 3, 2025

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